Extended Clique Models: A New Matching Strategy for Fingerprint Recognition

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Abstract

Establishing correspondences between minutiae sets is a fundamental issue in fingerprint recognition. This paper proposes a graph matching strategy. It addresses this problem using two extended clique models: the local clique model (LCM) and the global quasi-clique model (GQCM). First, minutia matching is formulated as maximal clique or quasi-clique detection. Similarities and compatibilities between minutia pairs are integrated and encoded into the correspondence graph. Correct matched minutia pairs then correspond to those strongly connected subgraphs, like the maximal clique or the maximal quasi-clique. Second, a two-level matching algorithm is proposed. In the local level, a novel similarity calculation of minutia topologic structures is proposed. It calculates similarities of minutia structures based on LCM and can be solved by maximal clique detection. In the global level, a novel similarity calculation of entire minutia sets is proposed. It calculates entire fingerprint similarity based on GQCM and can be solved by spectral correspondence techniques. Proposed method has stronger description ability and better robustness to noise and nonlinearity. Experiments conducted on FVC 2002 and 2004 databases demonstrate the effectiveness and the efficiency[1].

1. Introduction

Establishing correspondences between two feature sets is a fundamental issue in pattern recognition [1]. R. Horaud proposed a graph matching framework and formulated point matching as recovering the maximal cliques in the correspondence graph [2]. P.H. Tu introduced the maximal clique detection idea into latent fingerprint recognition [3]. However, recovering the maximal clique is time-consuming. Besides, as the existence of nonlinearity, correct minutia pairs may not form a clique in the correspondence graph. Clique is probably not a good constraint for distorted fingerprints.

It is challenging to find perfect correspondences for minutia sets due to various reasons: nonlinearity, noise and so on. Traditional approaches are to define some similarity measurements between two minutiae or two local minutia topologic structures (simplified as *LMTS*). Two strategies are widely used. One is to exact more matching features, such as minutiae phase difference features [4], minutiae ridge features [5]. The other is to construct complex *LMTS*s and define a local structural similarity for each minutia pair. A.K.J. proposed a structure composed by a minutia and a list of sample points on its corresponding ridge [6]. X. Chen constructed a topological structure to increase the tolerance for noise [7]. S. Chikkerur proposed k-plet structures and a graph-based representation [8]. W. Xu built local triangle structures [9]. R. Cappelli proposed a new representation called minutia cylinder-code [10]. These *LMTS*s take neighboring minutiae into consideration and are less impacted by nonlinearity and noise. However, the structural regions are not big. It limits the use of global information.

Based on these insights, we propose a novel graph matching strategy. As fingerprint images show both local rigidity and global nonlinearity, we propose two extended clique models: the local clique model (simplified as *LCM*) and the global quasi-clique model (simplified as *GQCM*), respectively. In the local level, a novel similarity measurement for *LMTS*s based on *LCM* is proposed. Calculating structural similarity is then formulated as recovering the maximal clique in the local correspondence graph [12, 13]. Minutia number within minutia structures is relatively small and the maximal clique detection time is acceptable. In the global level, a novel similarity measurement for entire minutia sets based on *GQCM* is proposed. Calculating entire similarity is then formulated as recovering the maximal quasi-clique in the global correspondence graph. We recover the maximal quasi-clique based on spectral techniques [14]. The spectral process converges after several iterations and the time is also acceptable.

Main contributions of this paper are: First, a novel graph matching strategy is proposed, minutia matching is formulated as maximal clique or quasi-clique detection. Second, a novel similarity calculation of *LMTS*s is proposed. It is based on *LCM* and can be solved by

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maximal clique detection. Third, a novel similarity measurement of entire minutia sets is proposed. It is based on *GQCM* and can be solved by spectral techniques.

2. Methodology

2.1. Problem formulation

We define an attributed graph as a graph $G_P = (V, E, A)$ for minutia set P. Each edge $e = ij \in E$ is assigned an attribute A_{ij}, which is the distance vector between minutia i and minutia j in P. We represent node attributes as special edge attributes, i.e. A_{ii} for node i. Let $G_P = (V, E, A)$ and $G_{P'} = (V', E', A')$ be two attributed graphs. Let $D_A(ii', jj')$ indicate the similarity score between attribute A_{ij} and $A'_{i'j'}$. We want to find a mapping that best preserves the attributes of nodes and edges between G_P and $G_{P'}$. Equivalently, we seek a set of minutia pairs $M = \{ii'\}$ so as to maximize the graph matching score, defined as:

$$\widetilde{M} = \underset{M}{\arg max} \sum_{ii', jj' \in M} D_A(ii', jj') \qquad (1)$$

Matching problems are usually formulated into maximum clique detection in correspondence graph [1]. The correspondence graph can be built, whose nodes represent similarities of minutia pairs and edges represent compatibilities between minutia pairs. Correct pairs are likely to establish links among each other and thus form a maximal clique. Incorrect minutia pairs establish links with the other pairs only accidentally, so they are unlikely to belong to strongly connected clusters. Matching problems are then formulated as recovering the maximal clique in the correspondence graph. However, traditional maximum clique model doesn't perform well for fingerprint matching, as it has nonlinearity in global regions and recovering maximal clique is time-consuming.

Here we make some improvements. We propose a *LCM* for local structural matching. Fingerprints can be seen as rigid within the range of minutia structures. Correct minutia pairs are likely to form the maximal clique in the local correspondence graph. The scale of the graph is small and the recovering process is very fast. Meanwhile, we propose a *GQCM* for entire minutia sets matching. Correct minutia pairs are likely to form the maximal quasi-clique in the global correspondence graph. Traditional clique detection algorithms are not suitable here for exponential time complexity. We use spectral techniques to get the approximate solution.

2.2. Local Clique Model

A typical *LMTS* is constructed from a center minutia and a list of neighboring minutia in a circular area. It can be seen as a small rigid fingerprint. Fingerprint images showed rigidity within the range of *LMTS*s, thus we can use a similar clique detection strategy to calculate their similarity.

Here we take *LMTS*s as the whole matching sets and the minutia pairs within structural regions as basic matching units. The local correspondence graph can be built, whose nodes represent similarities of minutia pairs and edges represent compatibilities between minutia pairs. Correct minutia pairs are likely to establish links among each other and thus they form a maximal clique. Incorrect pairs establish links with the other pairs accidentally, and they are unlikely to belong to strongly connected subgraphs. The main idea of *LCM* is shown in figure 1.

Nodes and edges with low values are useless, thus we set a similarity threshold and a compatibility threshold to prune. The local correspondence graph is very sparse and not big. By using clique detection algorithms [13], we can find the correct matched pairs and get the structural similarity.

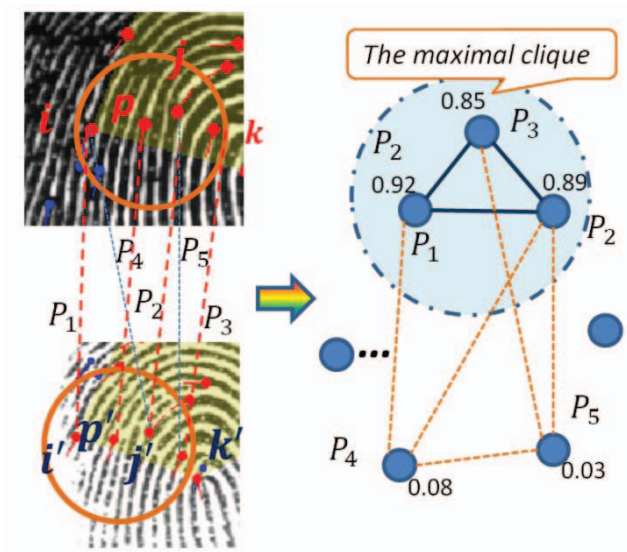

Figure 1: Illustration of the local clique model (*LCM*). Find correct minutia pairs within *LMTS*s and then form the local correspondence graph. Minutia matching corresponds to the maximal clique.

2.3. Global Quasi-Clique Model

Fingerprint images show nonlinearity in global regions. Correct matched pairs may not form a clique. Besides, the global correspondence graph is huge and the recovering process is inefficient. Here we propose *GQCM* which well solves these two problems.

Quasi-clique [15], [16] is a kind of dense subgraph in a weighted graph. The main properties of quasi-clique are internal homogeneity and external inhomogeneity. It means that the weights among internal nodes are overall higher than that between external and internal nodes. Compared to clique, quasi-clique usually misses a few edges. It is an extension of traditional clique. Here we take

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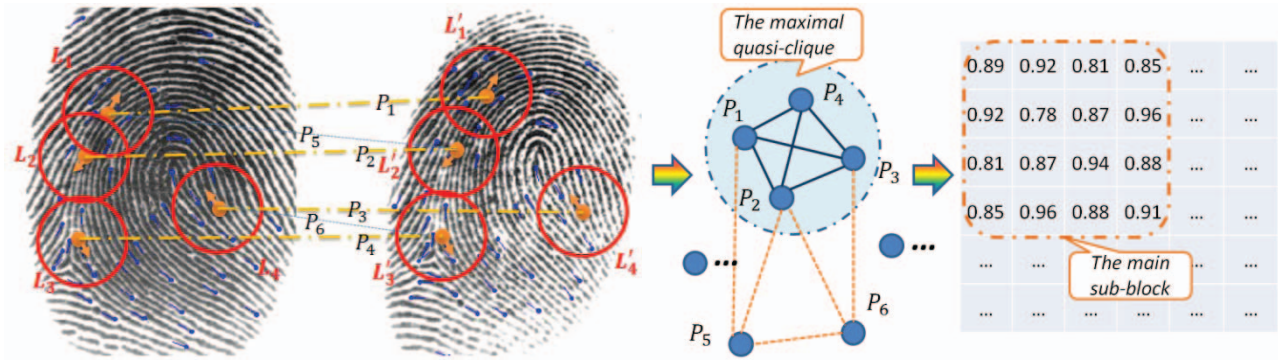

(a) (b) (c) (d)

Figure 2. Illustration of the global quasi-clique model (*GQCM*). Find all candidate *LMTS* pairs shown in (a) and (b) (for clarity, only a small subset of candidate pairs are shown), and then form the global correspondence graph in (c) and the global correspondence matrix in (d). Minutia matching corresponds to the maximal quasi-clique and the dense sub-block.

entire minutia sets as the whole matching parts and *LMTS*s pairs as basic matching units. The global correspondence graph can be built, whose nodes represent similarities of *LMTS* pairs and edges represent compatibilities between minutia pairs. Correct minutia pairs are likely to establish links among each other and thus they form a quasi-clique.

In order to improve the efficiency, we build the global correspondence matrix, which is another form of the local correspondence graph. Nodes are mapped into diagonal elements. Links are mapped into the non-diagonal elements. It is a symmetric matrix. Correct matched pairs can be recovered based on how strongly they belong to the main sub-block of the global correspondence matrix. The main idea of *GQCM* is shown in figure 2.

Formula 1 yields the following binary optimization problem:

$$\tilde{m} = arg \max_{m} (m^T D_A m), m \in \{0,1\}^{pq} \quad s.t. m\mathbf{1} \leq \mathbf{1} \; and \; m^T\mathbf{1} \leq \mathbf{1} \tag{2}$$

m is a row-wise vectored replica of D_A. The optimal solution $\tilde{m}$ is the binary vector that maximizes the $m^T D_A m$ score. It can be derived by using the principal eigenvector of D_A and imposing the one-to-one mapping constraints, which is proposed in [14]. Here we make some improvements. Firstly we prune the elements with small similarities or low compatibilities. Secondly we relax both the mapping constraints and the integral constraints on m, such that its elements can take real values in [0, 1]. Thirdly we don't initialize m^* randomly like [14], we use some priori knowledge. As minutia pairs with big local *LMTS* similarities are likely has large compatibilities, we initialize m^* by normalized similarities s_{nor} of *LMTS* pairs, shown in the next formula. It converges faster using this strategy.

$$m^*(i,j) = s_{nor}(i,j) = S_{ij} / \sum_i \sum_j {S_{ij}}^2$$

S_{ij} indicates the similarity between LMTS L_i and L_j. We can derive m^* by computing the leading eigenvalue of D_A, using the following algorithm.

Algorithm 1 Spectral iterations.

Input: matrix D_A
Output: m^*, main eigenvector of D_A

1 initial $m^* = s_{nor}$
2 repeat
3 ↓ $m^* \leftarrow D_A m^*$;
4 ↓ $m^* \leftarrow \frac{1}{||m^*||_2} m^*$;
5 until convergence;

Given the continuous solution m^*, we can rewrite it in a $N_p \times N_{p'}$ matrix form and discretize m^* by a greedy approach and derive the optimal solution $\tilde{m}$. Here we set the m^* elements whose value is smaller than a empirical threshold ϑ to zero. Meanwhile, minutia matching is one-to-one matching. We add this constraint during the iteration process. It is summarized as following:

Algorithm 2 Discretization.

Input: continuous matrix m^*
Output: binary matrix $\tilde{m}$

1 initialize $\tilde{m} = \mathbf{0}$
2 set $m^*(i,j) = 0, \forall i, \forall j, m^*(i,j) < \vartheta$
3 repeat
4 ↓ find the maximal element $m^*(i,j)$
5 ↓ set $\tilde{m}(i,j) = 1, \forall k$, set $m^*(i,k) = 0, m^*(k,j) = 0$
6 until $m^* = \mathbf{0}$;

After relaxation and discretization, we can gain the optimal matching *LMTS* pairs efficiently. Indeed, each step of the iteration algorithm requires only $O(z)$ operations, where z is the number of non-zero elements of the matrix. Also, typically, in our situation, the algorithm converges in a few dozen steps.

3. Fingerprint matching

3.1. Local Matching

A typical *LMTS* is constructed from a center minutia and a list of neighboring minutia in a circular area. Traditional methods use a longest common subsequence

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strategy to calculate the similarity of two *LMTSs* [7], [19]. It ignores the compatibility between each minutia pair within structural regions and it is time-consuming.

In the local level, *LMTS*s are the matching parts and the minutia pairs within *LMTSs* are basic matching units. As shown in figure 1, p and (i,j,k) form a minutia topologic structure, if we establish the polar coordinate system using the center point as center, coordinates of other points can be transformed into a distance vector $\bar{v}_{pi} = (d_{pi}, \alpha_{pi}, \beta_{pi})$. Variables $d_{pi}, \alpha_{pi}, \beta_{pi}$ indicate Euclidean distance, angle difference, coordinate angle difference between p and i, respectively.

Similarities of each minutia pair within *LMTSs* can be calculated according to the distance vectors. Take i and i' for example, the similarity $S_{ii'}$ can be calculated using the following formula.

$$S_{ii'} = \alpha \cdot e^{-\beta \cdot D(\bar{v}_{pi}, \bar{v}_{p'i'})} \qquad (3)$$
$$D(\bar{v}_{pi}, \bar{v}_{p'i'}) = ||\bar{v}_{pi} - \bar{v}_{p'i'}||_1$$

$D(,)$ measures the 1-norm distance of vectors $\bar{v}_{pi}$ and $\bar{v}_{p'i'}$. α, β are positive parameters.

Traditional compatibility describes the possibility of co-existence between two minutia pairs. Here we make some improvements. As each *LMTS* has a center minutia, which is also the origin of the polar coordinate system, we let the new compatibility describe the possibility of co-existence among the above two minutia pairs and the center minutia pair. Take (i,i') and (j,j') for example, the compatibility $C_{ii',jj'}$ can be calculated using the following formula:

$$C_{ii',jj'} = \alpha' \cdot e^{-\beta' \cdot D'(\bar{t}_{ijp}, \bar{t}_{i'j'p'})} \qquad (4)$$

$\bar{t}_{ijp}$ and $\bar{t}_{i'j'p'}$ are two triangles constituted by minutiae ijp and $i'j'p'$, respectively. $D'(\bar{t}_{ijp}, \bar{t}_{i'j'p'})$ measures their difference and can be calculated by adding $D(\bar{v}_{pi}, \bar{v}_{p'i'})$, $D(\bar{v}_{pj}, \bar{v}_{p'j'})$ and $D(\bar{v}_{ij}, \bar{v}_{i'j'})$. α', β' are positive parameters which are looser than the similarity parameters $\alpha, \beta (\alpha' > \alpha, \beta' < \beta)$.

After calculating similarities and compatibilities among minutia pairs, we make some pruning first. Candidate minutia pairs with similarities lower than the minimal threshold δ_s are removed. Suppose there is a *LMTS* pair $(L_p, L'_{p'})$ with n_p, $n_{p'}$ minutia, respectively. Because of the one-to-one mapping constraint, only subsets of size $n_s \le min\{n_p, n_{p'}\}$ can be matched. We retain $n_r = 3 * min\{n_p, n_{p'}\}$ candidate minutia pairs with the maximal similarity values. Next we build the local correspondence graph proposed in 2.2. It has only n_r nodes, which is much smaller than the original scale $(n_p * n_{p'})$. A compatibility threshold is set, too. Edges with compatibilities lower than δ_c is removed. The local correspondence graph is relatively small and very sparse.

We use the maximal clique detection algorithms for weighted graphs [13] to recover the maximal clique. Corrected matched pairs can be gained and similarities of *LMTS* pairs can be evaluated. Take (L, L') for example, the similarity $S_{LL'}$ is calculated using the following formula:

$$S_{LL'} = \sum_{ii' \in matched} S_{ii'} \cdot \frac{n_m^2}{n_p \cdot n_{p'}} \qquad (5)$$

$S_{ii'}$ measures the similarity of matched pairs ii'. $\sum_{i \in matched} S_{ii'}$ indicates the sum of matched pairs' similarities. n_m indicates the number of matched pairs within (L, L'), and n_p indicates the number within *LMTS* L, $n_{p'}$ has a similar meaning. This new measurement of *LMTSs* makes full use of local rigidity, which can be seen from the measurements of similarity and compatibility. It is more reasonable to adopt this strategy.

3.2. Global Matching

In the global level, the entire minutia sets are the matching parts and the *LMTS* pairs within it are basic matching units. Similarities and compatibilities can be calculated and the global correspondence graph can be constructed.

The *LMTSs'* similarities are calculated using the method proposed in 3.1. Note here, other strategies for measuring similarities of *LMTS*s could also be used, like the strategies proposed in [7], [10], [21]. Comparative experiments for different *LMTS* strategies are detailed in section 4.

Compatibility describes the possibility of co-existence between two *LMTS* pairs. Take $(L_p, L'_{p'})$ and $(L_q, L'_{q'})$ for example, as the minutia number within L_p and L_q is usually different, the *LMTS* distance vector $\bar{v}_{L_p L_q}$ does not have an explicit expression. We use the center minutiae's distance vector $\bar{v}_{pq}$ to approximate it.

$$C_{L_p L'_p, L_q L'_q} = \alpha^* \cdot e^{-\beta^* \cdot D'(\bar{v}_{pp'}, \bar{v}_{qq'})} \qquad (6)$$

$D'(,)$ measures the distance of two minutia distance vectors. α^*, β^* are positive parameters. As the existence of global nonlinearity, the global parameters α^*, β^* are much looser than the local parameters $\alpha', \beta' (\alpha^* > \alpha', \beta^* < \beta')$.

We make some pruning first. Candidate *LMTS* pairs with similarities lower than the minimal threshold φ_s are removed. Suppose there are two fingerprint minutia sets P and P', with N_p and $N_{p'}$ minutiae, respectively. N_p and $N_{p'}$ local minutia structures can be generated. Only subsets of size $N_s \le min\{N_p, N_{p'}\}$ can be matched for the one-to-one mapping constraint. We retain $N_r = 3 * min\{N_p, N_{p'}\}$ candidate minutia pairs with the maximal similarity values. Next we build the global correspondence matrix proposed in 2.3. It has only $N_r * N_r$ elements, which is much smaller than the original scale $(N_p * N_{p'}) *$

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$(N_p * N_{p'})$. It can be built using the following formula:

$$D_A(ii',jj') = \begin{cases} \lambda \cdot S_{L_iL_{i'}'} & if\ i = i'\ and\ j = j' \\ C_{L_iL_{i'}'L_jL_{j'}'} & else \end{cases} \quad (7)$$

The spectral iteration process is very fast. It usually will converge in several iterations. Finally the optimal minutia pairs can be gained through spectral techniques proposed in 2.3.

3.3. Regaining Minutia Pairs

Isolated minutiae or those minutiae in the border of fingerprint images have few surrounding minutiae. They may be removed at the step of constructing *LMTSs*. It is necessary to regain lost matches. We use the *moving least squares* transformation to regain lost genuine pairs [18]. For fingerprints from the same finger, many genuine minutiae can be retrieved. Meanwhile, for fingerprints from different fingers, few minutiae can be regained as the randomness of minutiae's position.

3.4. Scoring strategy using double convex hull

After gaining the matching relationship, we design a new scoring strategy to get the final matching scores. The score evaluation strategy based on convex hull can be represented as follows:

$$Score = \frac{\sum_{matched} S_{LL'}}{N_{\tilde{m}}} \cdot \frac{N_m^2}{(N_m+m_1)\cdot(N_m+m_2)} \quad (8)$$

$\sum_{matched} S_{LL'}$ indicates the sum of matched *LMTS*s' similarities. $N_{\tilde{m}}$ represents the number of initial matched minutia pairs gained in section 3.2. N_m represents the number of matched points after expanding, and m_1, m_2 represents the number of unmatched points within the convex hull. It is more reasonable to adopt this new score evaluation strategy.

4. Experiments and discussion

Experiments are conducted on FVC2002 and FVC2004 databases. We compare the proposed algorithm with some other well-known minutiae-only matching algorithms. Experimental results demonstrate the effectiveness of *LCM* and *GQCM*.

4.1. Validation of *Local Clique Model*

In order to test *LCM*'s ability of measuring local structural similarity, we do a special group of comparative experiments. Four different methods for measuring *LMTS* are used: *LTS* [7], *MCC* [10], *LSS* [21] and the proposed *LCM*. Here we only use the local matching part proposed in 3.1. The radius of minutia structures is set to 60 equally. After gaining the similarity matrix of *LMTS* pairs, we use the same discretization operation proposed in algorithm 2 to gain the optimal matching pairs. The final scores are evaluated using the same convex hull strategy, which is shown in the following formula:

$$Score = \sum_{matched} S_{LL'} \cdot \frac{N_m^2}{(N_m+m_1)\cdot(N_m+m_2)} \quad (9)$$

Results on FVC2002 and FVC2004 are shown in the Table 1. The *ROC* curves for FVC2004 DB1 are shown in figure 3. The experimental results show that our *LCM* algorithm is close to well-known *MCC*. It has promising performances in terms of both the EER and FMR.

Table 1 Comparative results on FVC2002 and FVC2004

EER	2002DB1	2002DB2	2004DB1	2004DB2
LTS[7]	2.68	3.34	8.16	10.81
LSS [21]	3.28	4.21	9.38	12.56
MCC[10]	**1.82**	2.47	6.25	**8.54**
LCM	2.25	**1.89**	**5.36**	8.97

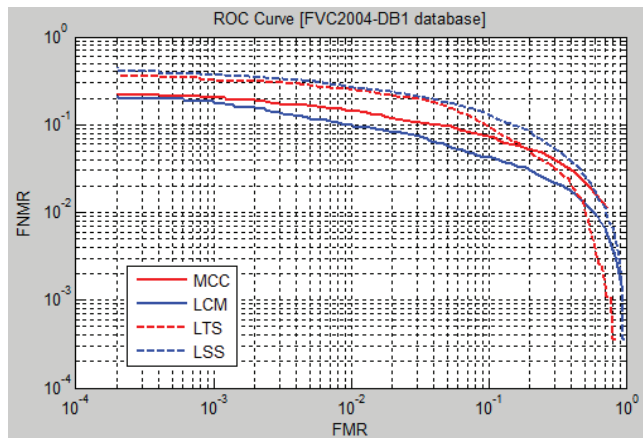

Figure 3. ROC curves for local matching on FVC2004 db1

4.2. Validation of *Global Quasi-clique Model*

MCC [10] and the proposed *LCM* are two better methods for measuring similarities of *LMTSs* pairs. They can be used to calculate the structural similarity in the global matching. Take *GQCM* into consideration, we design two combined algorithms: *MCC+GQCM*, and *LCM+GQCM*. We compare these two algorithms with some other algorithms, including [9] and [11]. Results are shown in the Table 2. The corresponding *ROC* curves for FVC2004 DB1 are shown in figure 4. The results show that the algorithms based on *GQCM* outperforms the other algorithms.

5. Conclusion

This paper introduces a graph matching strategy by proposing two extended clique models: the local clique model and the global quasi-clique model. In the local level, a novel measurement for *LMTS*s is proposed. It makes fully use of local rigidity. Minutia structural matching is formulated as recovering the maximal clique in the local

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Table 2 Comparative results on FVC2002 and FVC2004: The best results are marked by bold face.

DataBase	2002 DB1		2002 DB2		2004 DB1		2004 DB2	
	EER	FMR1000	EER	FMR1000	EER	FMR1000	EER	FMR1000
SC[11]	2.54	5.14	3.76	8.46	9.02	17.69	10.31	25.24
GrowFuse [9]	1.08	1.27	1.32	1.94	4.21	8.824	4.91	9.21
MCC+ GQCM	**0.42**	**0.69**	0.85	1.95	3.77	9.12	5.94	11.13
LCM+GQCM	0.57	0.98	**0.71**	**1.92**	**3.13**	**7.54**	**3.625**	**7.914**

correspondence graph. It can be solved by clique detection algorithms efficiently. In the global level, a novel measurement for entire minutia sets is proposed. It takes nonlinearity into consideration. Minutia matching is formulated as recovering the maximal quasi-clique in the global correspondence graph. It can be solved by spectral techniques efficiently. It has stronger description ability and better robustness to nonlinearity and noise. Experimental results demonstrate the effectively and accuracy of our algorithm.

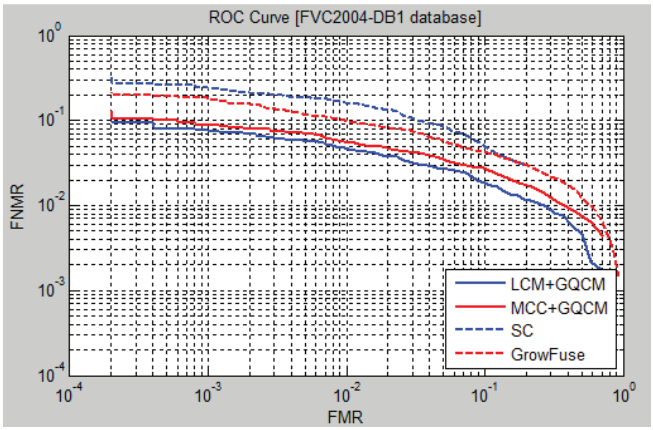

Figure 4: ROC curves on FVC2004 db1

References

[1] D. Conte, P. Foggia, C. Sansone, M. Vento. Thirty years of graph matching in pattern recognition. International Journal of Pattern Recognition and Artificial Intelligence. 18(3): 265–298, 2004

[2] R. Horaud, T. Skordas, Stereo correspondence through feature grouping and maximal cliques. IEEE Transactions on Pattern Analysis and Machine Intelligence vol.11: 1168-1180, 1989.

[3] Peter Henry Tu, Richard Ian Hartley. Fingerprint matching by estimation of a maximum clique. Number US Patent 5933516, 1997

[4] H. Choi, K. Choi, and J. Kim, Fingerprint Matching Incorporating Ridge Features With Minutiae, Transactions on Information Forensics and Security. Vol 6, No 2, 2011.

[5] C. Liu, J. Cao, X. Gao, A novel fingerprint matching algorithm using Minutiae Phase Difference Feature, International Conference on Image Processing, 2011.

[6] A.K. Jain, L. Hong, R. Bolle. On-Line Fingerprint Verification. IEEE Transactions on Pattern Analysis and Machine Intelligence, 19(4): 302–314, 1997.

[7] X. Chen, J. Tian, X. Yang, A Matching Algorithm Based on Local Topologic Structure, International Conference on Image Analysis and Recognition, LNCS 3211, pp. 360-367 2004

[8] S. Chikkerur, A.N. Cartwright, V. Govindaraju. K-plet and Coupled BFS: A Graph Based Fingerprint Representation and Matching Algorithm, International Conference on Biometrics, pp. 309–315, 2006.

[9] W. Xu, X. Chen, J. Feng. A Robust Fingerprint Matching Approach: Growing and Fusing of Local Structures. International Conference on Biometrics, LNCS 4642, 134-143, 2007

[10] R. Cappelli, M. Ferrara, D. Maltoni, Minutia Cylinder-Code: a new representation and matching technique for fingerprint recognition. IEEE Transactions on Pattern Analysis and Machine Intelligence, 32(12): 2128-2141, 2010.

[11] Y. Feng, J. Feng, X. Chen, Z. Song, A Novel Fingerprint Matching Scheme Based on Local Structure Compatibility, International Conference on Pattern Recognition, vol 4, pp 374-377, 2006

[12] PRJ. Östergård, A fast algorithm for the maximum clique problem," Discrete Applied Mathematics, 120: 195-205, 2002.

[13] PRJ. Östergård, A new algorithm for the maximum-weight clique problem, Electronic Notes in Discrete Mathematics, vol. 3, pp. 153-156, 1999

[14] M. Leordeanu and M. Hebert, A Spectral Technique for Correspondence Problems Using Pairwise Constraints, Int'l Conf. of Computer Vision, vol. 2, pp. 1482-1489, 2005.

[15] G. Liu and L. Wong, Effective Pruning Techniques for Mining Quasi-Cliques, Machine Learning and Knowledge Discovery in Databases Lecture Notes in Computer Science vol 5212, , pp 33-49, 2008

[16] M. Pavan, A New Graph-Theoretic Approach to Clustering, with Applications to Computer Vision, Ph.D. Thesis, 2004

[17] J. Weibull. Evolutionary game theory. MIT Press,1997.

[18] S. Schaefer, T. McPhail, J. Warren, "Image Deformation Using Moving Least Squares," ACM Transactions on Graphics, vol. 25, issue 3, pp. 533-540, 2006.

[19] J. Cao, J. Feng, A Robust Fingerprint Matching Algorithm Based on Compatibility of Star Structures. In MIPPR, vol. 7498, pp.74983X, 2009

[20] T.Y. Jea, V. Govindaraju. A minutia-based partial fingerprint recognition system. Pattern Recognition, 38(10): 1672–1684, 2005

[21] N.K. Ratha, V.D. Pandit, R.M. Bolle, and V. Vaish, "Robust Fingerprint Authentication Using Local Structural Similarity," Applications of Computer Vision, IEEE Workshop on, p. 29, 2000.

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